

Lecture - 7

Theorem Let A be an $n \times n$ matrix. The following statements are equivalent

- ① A is invertible
- ② The reduced row echelon form of A is I_n

Proof

② $\Rightarrow$ ① If the row reduced echelon form of A is I_n then there is a matrix B such that $BA = I_n$.

B is a product of elementary matrices
Therefore B is invertible.
See Tutorial - 4, Problem 5

Then, we know that $AB = I_n$ as well. Thus A has an inverse, namely B .

① $\Rightarrow$ ② Suppose that A is invertible, say, $BA = I_n = AB$ and the row reduced echelon form of A is J . That is, there is a matrix C such that $CA = J$.

Suppose that J is NOT the identity matrix. Then, J must have zero rows. Therefore J cannot be invertible.

Since C is a Product of elementary matrices and that elementary matrices are invertible, we obtain that C is invertible as well. Now since A is invertible, it follows that $CA = J$ is invertible. A Contradiction!

This prove that the Statement

"Suppose that J is NOT the identity matrix."
is incorrect. That is, the row echelon form J must be the identity matrix I_n .

Calculating inverse of a matrix

- (a) Inverse of A need not exists! $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
- (b) If the inverse exists then it is unique.

What we have observed earlier is that

- (c) "A has an inverse" is the same as "the reduced row echelon form of A is I_n ".

A method for calculating inverse (if it exists)

- If $B_1, B_2, \dots, B_n$ are elementary matrices such that $B_1 B_2 \dots B_n A = I_n$ then $B = B_1 B_2 \dots B_n$ is the inverse of A .

- Consider the augmented matrix

$$\begin{array}{ccccc} \left(A \mid I_n \right) & \xrightarrow{\quad} & \left(B_n A \mid B_n \right) & \xrightarrow{\quad} & \left(B_{n-1} B_n A \mid B_{n-1} B_n \right) \\ & & & & \downarrow \vdots \\ \left(I_n \mid B \right) & \xleftarrow{\quad} & \left(B_1 \cdots B_n A \mid B_1 B_2 \cdots B_n \right) & \xleftarrow{\quad} & \end{array}$$

inverse!
reduced row echelon form

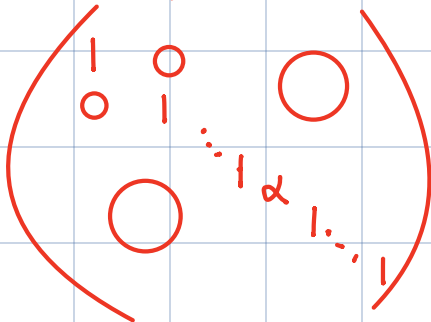
Elementary matrices

Let E_{ij} be the matrix whose ij^{th} position is 1 and 0 everywhere else.

The following kinds of matrices are called elementary matrices.

① $E_{11} + E_{22} + \dots + E_{i-1,i-1} + \alpha E_{ii} + E_{i+1,i+1} + \dots + E_{nn}$

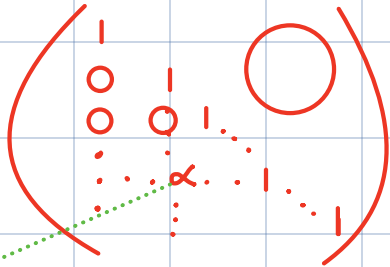
This matrix corresponds to $R_i \rightarrow \alpha R_i$



Same as $I_n + (\alpha - 1) E_{ii}$

② $I_n + \alpha E_{ij} ; i > j$

This matrix corresponds to $R_i \rightarrow R_i + \alpha R_j$



i^{th} entry

③ Switching of rows R_i and R_j

$I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$, which is the matrix obtained by switching the i^{th} and j^{th} rows of the identity matrix I_n

Example

Find the inverse of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $ad-bc \neq 0$ and $a \neq 0$, using row reduction method

Solution.

$$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{a} R_1} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - c R_1} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & d - \frac{bc}{a} & -\frac{c}{a} & 1 \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & d - \frac{bc}{a} & -\frac{c}{a} & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow \frac{a}{ad-bc} R_2} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 - \frac{b}{a} R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right)$$

Thus the inverse of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is

$$\begin{pmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$$

HW

If $a=0$
then $b \neq 0$
as $ad-bc \neq 0$.

Write Solution
for this Case!

Example Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc = 0$. Then A has no inverse!

First we assume $a \neq 0$. Then

$$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{a}R_1} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - cR_1} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 0 & -\frac{c}{a} & 1 \end{array} \right)$$

Thus A has no inverse!

H.W

If $a = 0$ then $b = 0$ or $c = 0$. In these cases show that A has no inverse!